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Chapter 13 Securities Analysis: Doing a Little Market Research

In terms of choosing securities, throwing darts at a list of stocks seems to have fallen out of favor. So has drawing company names out of a hat. But, hey, no problem. Your psychic powers may not be the most reliable, but you still have tons of tools that can help you get a good idea of where the market's heading and how certain securities may perform.

One of your main jobs as a registered representative is to figure out the best investments for your customers based on their investment objectives and your research. To help lead people down the path to riches, you have to analyze each customer's portfolio and the market and try to find a good fit. In many cases, firms hire analysts to provide registered reps investment information, which helps you (as the registered rep) determine the best recommendations for each customer.

In this chapter, I cover topics relating to securities analysis and money supply. The majority of this chapter is about analyzing companies and the market and seeing what happens with the money supply. Don't worry, though — I don't leave out technical and fundamental analysis. I just focus on the information that can help you get the best score on the SIE. At the end, you get to test your knowledge with a quick chapter quiz.

Getting to Know Your Securities and Markets: Securities Analysis Basics

Although many brokerage firms have their own analysts, you do need to know some of the basics of securities analysis to pass the SIE and corequisite exams. Besides, the more you know about securities analysis, the better you'll be able to understand the analysts and the more informed you'll sound when talking to your customers and potential customers. In this section, I cover investment risks that your customers face and show you the differences between technical and fundamental analysis.

Regarding systematic and nonsystematic risk

Investors face many risks (and many rewards) when investing in the market. You need to understand the risks because this knowledge cannot only make you sound like a genius, but also help you score higher on the SIE exam.

Systematic risk

Systematic (undiversifiable or market) risk is the risk that securities can decline due to political, social, or economic factors — changes in the economy, natural disasters, government policy, and so on. Examples of systematic risks are the housing crisis of 2008 and COVID-19, which started in 2020 and is still negatively affecting our economy. Systematic risk is a risk that could affect the whole market. Systematic risks include the following:

- **Market risk:** The risk of a security or securities declining due to regular market fluctuations or negative market conditions. All securities have market risk.
- **Interest rate risk:** The risk of bond prices declining with increasing interest rates. (Use the idea behind the seesaw from Chapter 7: When interest rates increase, outstanding bond prices decrease.) All bonds, even zero-coupon bonds, are subject to interest risk.
- **Reinvestment risk:** The risk that interest and dividends received will have to be reinvested at a lower rate of return; zero-coupon bonds, T-bills, T-STRIPS, and so on have no interperiod reinvestment risk (until maturity) because they don't receive interest payments.
- **Purchasing power (inflation or inflationary) risk:** The risk that the return on the investment is less than the inflation rate. As of the time of this writing, we're experiencing the worst inflation in more than 40 years. Long-term bonds (even Treasury bonds) and fixed annuities have high inflation risk. To avoid inflation risk, investors should buy stocks and variable annuities.

Nonsystematic risk

Nonsystematic (unsystematic, unique, or diversifiable) risk is more industry- or firm-specific. The good news is that this type of risk can be eliminated through diversification. You've probably heard the expression "Don't put all your eggs in one basket." Well, the same holds true for investing. Suppose that one of your customers has everything, or a large portion of their investment money, invested in DIMP Corporation common stock, and DIMP files for bankruptcy; now you have to tell your customer they've lost all (or at least a large portion of) their investment money.

- **Business risk:** The risk of a corporation failing to perform up to expectations.
- **Political (geopolitical) risk:** The risk that the value of a security could suffer due to instability or political changes in a country (such as the nationalization of corporations).
- **Default risk (Credit risk):** The risk of default or that the principal and interest aren't paid on time; Moody's, Standard & Poor's, and Fitch are the main bond-rating companies.
- **Regulatory risk:** The risk that changes in the regulatory climate (rulings by the Food and Drug Administration, Environmental Protection Agency, and so on) will have a detrimental affect on certain securities in the market.

- **Legislative risk:** The risk that changes in state or federal law will affect certain securities in the market.
- **Currency (exchange rate) risk:** The risk that an investment's value will be affected by a change in currency exchange rates; long-term investors who have international investments are the ones most affected by currency risk.

  REMEMBER

  Many investors buy and sell currencies in an attempt to take advantage of currency exchange rates. The initial currency to be traded (the base currency) would be exchanged for another currency (the counter currency). To determine the amount of counter currency they would receive, they would look at the spot exchange rate.
- **Liquidity (marketability) risk:** The risk that the security is not easily traded without affecting the price of the security; long-term bonds and limited partnerships have more liquidity risk.
- **Capital risk:** The risk of losing all money invested (for options [Chapter 11] and warrants [Chapter 6]); because options and warrants have expiration dates, purchasers may lose all money invested at expiration. To reduce capital risk, investors should buy high-quality stocks or investment-grade (higher rated) bonds.
- **Prepayment risk:** The type of risk mostly associated with real-estate investments such as mortgage-backed securities (Chapter 7); mortgage-backed securities have an average expected life when first issued, but if mortgage interest rates decrease, more investors will refinance, and the bonds will be pre-paid earlier than expected.
- **Timing risk:** The risk of an investor buying or selling a security at the wrong time, thus failing to maximize profits.

TIP

When taking the real exam and practice exams, you should always pay close attention to the investor's risk tolerance, financial considerations, nonfinancial considerations, and risk(s) mentioned in order to determine the best investment for a particular customer.

Strategies for mitigating risk

Certainly, all investments have a certain degree of risk. Younger investors, sophisticated investors, and wealthy investors can all afford to take more risk than the average investor. However, when you are talking to your clients, you should help them make decisions that will help them mitigate their risk. Those in charge of coming up with questions for the SIE exam would expect you to be able to navigate the whole risk topic effectively, so you can be sure that some questions will address the following topics directly.

Diversification

Consider again the customer mentioned in the earlier section "Nonsystematic risk" who has everything invested in DIMP Corporation common stock. All of a sudden, DIMP Corporation loses a big contract or is being investigated. Your customer could be wiped out. However, if your customer had a diversified portfolio, DIMP Corporation would likely be only a small part of their investments, and they wouldn't be ruined. This is the reason that having a diversified portfolio is so important.

There are many ways to diversify, including the following:

- **Geographical:** Investing in securities in different parts of the country or world.
- **Buying bonds with different maturity dates:** Buying a mixture of short-term, intermediate-term, and long-term debt securities.

- **Buying bonds with different credit ratings:** Purchasing high-yield bonds (bonds with a low credit rating, also known as *junk bonds*) in combination with highly rated bonds with lower returns so that you get a mixture of high returns with the safety of the highly rated bonds.
- **Investing in stocks from different sectors:** Assumingthat different sectors may perform better than others at certain times, it definitely makes sense to look at different sectors when considering which securities to buy. By spreading your investments out among these different sectors, you can manage your risk and ideally make a profit if one or more sectors happen to be performing well. Sectors include financials, utilities, energy, healthcare, industrials, technology, and so on.
- **Type of investment:** Investing in a mixture of different types of stocks, bonds, direct participation programs (DPPs), real estate, options, and so on.

TIP

There are certainly many more ways to diversify a portfolio than the ones listed previously; use your imagination. In addition, they aren't mutually exclusive. Remember that mutual funds (packaged securities) and exchange traded funds (ETFs) provide a certain amount of diversification within an individual holding. This is why smaller investors who may not be able to afford to diversify their portfolio are ideal candidates for mutual funds.

Portfolio rebalancing

Say that you and one of your clients determine that it is best for them to have a portfolio of 50 percent equity securities and 50 percent debt securities. After setting up and purchasing the portfolio, one year later, due to appreciation, your client has 60 percent in equity securities and 40 percent in debt securities. At that point, your client may decide to rebalance their portfolio by selling some equity securities and purchasing more debt securities to help maintain their original desired level of asset allocation (50–50). As a matter of fact, a subset of mutual funds known as *asset allocation funds* will rebalance the portfolio of securities held by the fund without needing to contact the shareholders.

REMEMBER

Typically, as investors age, they can't afford to take as much risk and should change their asset allocation to include fewer equity securities and more debt securities.

Hedging

I'm sure you've heard the saying "Hedge your bets." In the gamblers' world, this may mean taking insurance to protect oneself against the possibility of the dealer getting a 21 when playing blackjack. It means that you are trying to reduce your risk. The problem is that by doing so, you may be limiting your upside potential. In the investors' world, there are several ways to hedge, depending on what you're investing in.

You can hedge (protect) your investments against market volatility (the risk that the market will fluctuate in price) by having a diversified portfolio. In this case, although some of your investments may be subject to big swings in price, others will remain stable or increase when the market decreases. A well-diversified portfolio may include short- and long-term bonds of varying credit risk, cash equivalents (in other words, money market funds), all sorts of equity securities (value, growth, large-cap, small-cap, and so on), real-estate investments, commodities, precious metals, and so on.

To hedge against credit risk (the risk that bond issuers will default), you can purchase some more secure debt securities issued by the U.S. government (T-bills, T-notes, T-bonds, and so forth), debt securities issued by local governments (municipal bonds and municipal notes), debt securities by higher-rated corporations, and so on.

To hedge against the risk that a security doesn't keep pace with inflation, you can purchase stocks (best answer in for an SIE question), variable annuities, real estate, commodities (raw materials or agricultural products), or even Treasury Inflation-Protected Securities (TIPS).

REMEMBER

Just as there are many types of risk (as you can see in the previous sections "Systematic risk" and "Nonsystematic risk"), there are different ways to hedge against risk, depending on which risk you're concerned about. The preceding are just a few examples. Options can also be used to hedge against the risk of a security you own going in the wrong direction. The best way to limit risk for most investors is to have a diversified portfolio. You must also understand that in most cases, older investors can't afford to take as much risk. For SIE exam purposes, the main thing you need to remember is that *hedge* means to protect.

Deciding what to buy: Fundamental analysis

Although most analysts use some combination of fundamental analysis and technical analysis to make their securities recommendations, for SIE exam purposes, you need to be able to differentiate between the two types. This section discusses fundamental analysis; I cover technical analysis later in the section "Deciding when to buy: Technical analysis."

Fundamental analysts perform an in-depth analysis of companies. They look at the management of a company and its financial condition (balance sheets, income statements, the industry, management, earnings, and so on) and compare with other companies in the same industry. They can also compare many years of financial statements to help determine whether a company is heading in the right direction. In addition, fundamental analysts look at the overall economy and industry conditions to determine whether an investment is good to buy.

REMEMBER

In simplest terms, fundamental analysts decide *what to buy.*

A fundamental analyst's goal is to determine the value of a particular security and decide whether it's underpriced or overpriced. If they believe the security is underpriced, a fundamental analyst recommends buying the security; if they believe the security is overpriced, they recommend selling outright or selling the security short.

The following sections explain some of the fundamental analyst's tools of the trade and how to use them.

Balance sheet components

The *balance sheet* provides an image of a company's financial position at a given point in time. The SIE exam tests your ability to understand the components of a balance sheet (see Figure 13-1) and how financial moves that the company makes (buying equipment, issuing stock, issuing bonds, paying off bonds, and so on) affect the balance sheet. In general, understanding how a balance sheet works is more important than being able to name all the components.

REMEMBER

People call this statement a balance sheet because the assets must always balance out the liabilities plus the stockholders' equity.

Assets	Liabilities
Current assets Fixed assets Intangible assets	Current liabilities Long-term liabilities
	Stockholder's equity (net worth)
	Par value (common) Par value (preferred) Paid-in capital Treasury stock Retained earnings

FIGURE 13-1: Components of a balance sheet.

Assets are items that a company owns. They include

- **Current assets:** Owned items that are easily converted into cash within the next 12 months; included in current assets are cash, marketable securities, accounts receivable, inventory, and any prepaid expenses (like rent or advertising).

  Note: Fundamental analysts also look at methods of inventory valuation, such as *LIFO* (last-in-first-out) or *FIFO* (first-in-first-out). In addition, they look at the methods of depreciation, which are either *straight line* (depreciating an equal amount each year) or *accelerated* (depreciating more in earlier years and less in later years).
- **Fixed assets:** Owned items that aren't easily converted into cash; included are property, building(s), furniture, and equipment. Because many fixed assets wear down or become outdated over time, they can be depreciated (except for land). Therefore, accumulated depreciation is usually deducted from fixed assets.
- **Intangible assets:** Owned items that don't have any physical properties; included are items such as trademarks, patents, formulas, copyrights, goodwill, and so on. (Created when a corporation purchases or merges with another company, *goodwill* is the dollar amount paid above the fair market value to purchase that company.)

Liabilities are what a company owes. They may be current or long-term:

- **Current liabilities:** Debt obligations that are due to be paid within the next 12 months; included in current liabilities are *accounts payable* (what a company owes in bills), wages, debt securities due to mature, short-term *notes payable* (the balance due on money borrowed), declared cash dividends, and taxes.
- **Long-term liabilities:** Debt obligation due to be paid after 12 months; included in long-term liabilities are mortgages, bank loans, outstanding corporate bonds, and long-term notes.

Stockholders' equity (net worth) is the difference between the assets and the liabilities (basically, what the company is worth). This value includes

- **Par value of the common stock:** The arbitrary amount that the company uses for bookkeeping purposes; if a company issues 1 million shares of common stock with a par value of $1, the par value on the stockholders' equity portion of the balance sheet is $1 million.
- **Par value of the preferred stock:** The value that the company uses for bookkeeping purposes (usually $100 per share but could be $25, $50, $1,000, or some other number); if the company issues 10,000 shares of preferred stock at $100 par, the par value of the stockholders' equity portion of the balance sheet is $1 million.

REMEMBER

Unlike common stock, preferred stock has a par value that is typically $100 and a stated dividend rate; preferred stock shareholders would also receive money prior to common stockholders in the event of corporate bankruptcy.

- **Additional paid-in capital:** The amount over par value that the company receives for issuing stock; if the par value of the common stock is $1 but the company receives $7 per share, the additional paid-in capital is $6 per share. The same theory holds true for the preferred stock.
- **Treasury stock:** Stock that was outstanding in the market but was repurchased by the company.
- **Retained earnings:** The amount of net earnings the company holds after paying out dividends (if any) to its shareholders.

Income statement components

An income statement tells you how profitable a company is currently. *Income statements* list a corporation's expenses and revenue for a specific period of time (quarterly, year-to-date, or yearly). When comparing revenue and expenses, you should be able to see the efficiency of the company and how profitable it is.

Again, I don't think you need to actually see a detailed income statement from a company, but knowing the components of an income statement is important. Take a look at Figure 13-2 to see how an income statement is laid out. Most of the items are self-explanatory.

Net sales
\- Cost of goods sold (earnings before interest, taxes, depreciation, and amortization) (EBITDA)
\- Operating expenses (including depreciation)
Operating profit (earnings before interest and taxes) (EBIT)
\- Interest expenses
Taxable income (earnings before taxes) (EBT)
\- Taxes
Net income (earnings after taxes) (EAT)
\- Preferred dividends
Earnings available to common stockholders
\- Common dividends
Retained earnings

FIGURE 13-2: Components of an income statement.

Deciding when to buy: Technical analysis

Technical analysts look at the market to identify patterns and measure indicators in an attempt to predict whether the market and/or particular securities will become or remain bullish or bearish. They look at trend lines, trading volume, market sentiment, market indices (Standard & Poor's [S&P] 500, Dow Jones Industrial Average [DJIA], and so on), options volatility, market momentum, available funds, index futures, new highs and lows, the advance-decline ratio, odd lot volume, short interest, put-to-call ratio (options trading), and so on. These analysts believe that history tends to repeat itself and that past performance of securities and the market indicate its future performance.

REMEMBER

Fundamental analysts decide *what to buy*, and technical analysts decide *when to buy* (timing).

Technical analysts chart not only the market, but also market sectors and individual securities. Technical analysts try to identify market patterns and patterns of particular securities in an attempt to determine the best time to purchase or sell. Even though a security's price may vary a lot from one day to another, the prices tend to head in a particular direction (up, down, or sideways) and create a *trend line* over a period of time.

Benchmarks and indices

If you watch news stations, read the newspaper, listen to the radio, and so on, you can't help but see or hear about the DJIA or the Nasdaq being up or down. Well, those are indices (indexes) or benchmarks. Benchmarks are typically used to evaluate the performance of individual investments or a group of investments. Most investors compare their investments with certain broad-based or narrow-based indices:

- **Narrow-based:** Narrow-based indices indicate the performance of a particular industry such as the Dow Jones Transportation Index.
- **Broad-based:** Broad-based indices are more indicative of the overall market. Broad-based indices measure securities from many industries.

TIP

There are certainly more indices than the ones listed, but for SIE exam purposes, you shouldn't need to memorize them — mainly understand what indices are and that they are often used as benchmarks.

Here are examples of some of the broad-based stock indices:

- **S&P 500 Index:** Includes 500 large-cap (companies that have a market capitalization above $10 billion) common stocks.
- **Wilshire 5000 Total Market Index:** The largest of all stock indexes; includes 5,000 listed common stocks.
- **Russell 2000 Index:** An index of 2,000 small-cap (companies that have a market capitalization between $300 million and $2 billion) companies.
- **Lipper Indexes:** Track the financial performance of different mutual funds based on their investment strategy; each Lipper Index tracks the performance of only the largest fund in each category (large-cap growth, mid-cap value, international fund, and so on).
- **Dow Jones Composite Average:** An index that tracks 65 stocks from some of the most prominent companies; the Dow Jones Composite is broken into
  - *Dow Jones Industrial Average (DJIA):* Tracks 30 stocks from the industrial sector; the DJIA is the index most commonly used to indicate the performance of the market in general.
  - *Dow Jones Transportation Average:* Tracks 20 stocks from the transportation sector
  - *Dow Jones Utility Average:* Tracks 15 stocks from the utility sector

TIP

Proponents of the *Dow Theory* believe that major market trends are confirmed if the Dow Jones Industrial Average and the Dow Jones Transportation Average are trending in the same direction (that is, both advancing or both declining). Logic dictates that if industrial companies are producing more goods, those same goods need to be transported.

Most of the indices listed here are weighted toward larger companies. This means that price movement of the larger companies has a greater impact on the particular index than a smaller company does.

Stages of the business cycle

The business cycle is the natural rise and fall of goods and services (gross domestic product, or GDP) that occur over time. The business cycle has four phases that will occur over and over:

- **Expansion** (*A* in Figure 13-3): *Expansion* is characterized by increasing demand for goods and services. During expansion, the stock market is generally increasing (bullish), property values are increasing, and industrial production is increasing. Expansion also can be characterized as recovery.
- **Peak** (*B* in Figure 13-3): The *peak* occurs at the top of the expansion phase and happens right before the economy starts to contract.
- **Contraction** (*C* in Figure 13-3): *Contraction* is characterized by higher levels of consumer debt, a stock market that is generally decreasing (bearish), a decreasing demand for goods and services, and an increasing number of bond defaults and bankruptcies.
- **Trough** (*D* in Figure 13-3): *Trough* is the lowest part of the contraction phase and happens right before the economy starts to expand (recover) again.

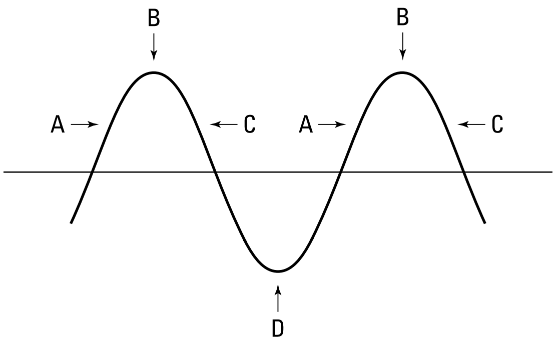

FIGURE 13-3: The four phases of the business cycle.

TIP

If asked to place these phases in order on the SIE exam, you can put them in order just as they're given in the preceding list.

Bullish versus bearish

When thinking of whether the market is bullish or bearish, think of the terms. You can think of bullish as charging ahead. So, if the market is bullish, it is generally increasing in value. If the market is bearish, it is generally hibernating or sleeping. When the market is bearish, it is generally decreasing in value.

Individuals can be bullish or bearish on the market in general or bullish or bearish on certain securities.

- **Bullish** strategies include buying individual stocks, buying mutual funds, buying call options, selling uncovered (naked) put options, and so on.
- **Bearish** strategies include selling short individual stocks, buying bearish funds (funds that generally increase in value in a declining market), buying inverse exchange-traded funds (ETFs), selling uncovered (naked) call options, buying put options, and so on.

Following the Green: Money Supply and Monetary Policy

The money supply heavily affects the market. If the money supply is higher than average, typically interest rates go down, people usually borrow more money, and people spend more money. That all sounds great, but the situation can lead to some negatives, such as higher inflation and the weakening of U.S. currency in relation to foreign currency. The Federal Reserve Board (the Fed or FRB) tries to do a balancing act to help the economy grow at a slow and steady rate. This section deals with how the money supply affects the market and the tools that the Fed uses to control the money supply.

REMEMBER

The Fed controls the monetary policy, but the *fiscal policy* is controlled by government politicians (the House, the Senate, and ultimately signed by the president). The fiscal policy is typically included in budget decisions and includes how much the U.S. government will borrow (and how), how much it will spend (and on what), how much money will be raised through taxes, and so on.

TIP

To put it in a nutshell, so to speak, you can think of monetary and fiscal policies like this:

- Monetary policy = money supply, interest rates
- Fiscal policy = borrowing, spending, taxes

Influencing the money supply

Changes in money supply can affect rates of economic growth, inflation, and foreign exchange, so knowing a bit about monetary policy can help you predict how certain securities will fare and how interest rates will change. Take a look at Table 13-1 to see what easing and tightening the money supply can do.

TABLE 13-1 Effects of Easing and Tightening the Money Supply

Category	Easing the Money Supply	Tightening the Money Supply
Economy	Easy money helps the United States avoid or get out of a recession. Consumers can borrow money at lower interest rates.	High interest rates slow the economy down because people aren't spending and investing as much money; the rate of small-business failure increases.
Market	As a result of lower interest rates, investors have more money to invest and can purchase more goods. Additionally, businesses don't have to pay as much interest to borrow money, which increases their profits. Both elements can lead to a bullish market.	High interest rates hurt the market because investors don't have extra money to invest. Additionally, corporations have to pay higher interest on loans and, therefore, report lower earnings. The market becomes bearish.
Inflation	Lower interest rates lead to higher inflation. If companies see that customers are spending money freely, they raise their prices.	A tighter money supply helps curb high inflation.
Strength of the U.S. dollar	The U.S. dollar weakens. U.S. exports increase because foreign currency strengthens (people can trade fewer units of foreign currency for more dollars); therefore, buying U.S. products is cheaper for foreign consumers. However, the U.S. dollar loses value for purchasing foreign goods, so foreign imports decrease.	The value of the U.S. dollar rises in relation to foreign currency. The U.S. dollar is subject to supply and demand, so if our money supply is tight, the value of our currency increases. Because the U.S. dollar is strong, importing foreign goods is cheaper for U.S. companies. However, U.S. exports decline because buying U.S. goods becomes more expensive for foreign companies.

When the money supply is eased (resulting in *easy money*), interest rates in general decrease. The Fed can ease the money supply by

- Buying U.S. government securities in the open market
- Lowering the discount rate, reserve requirements, and/or Regulation T (although changing Reg T isn't likely)
- Printing U.S. currency

Occasionally, the Fed has to tighten the money supply. (Remember that the Fed wants the U.S. economy to grow at a slow, steady pace.) When the money supply is tightened (resulting in *tight money*), interest rates across the board increase. The Fed can tighten the money supply by

- Selling U.S. government securities (pulling money out of the banking system)
- Increasing the discount rate, increasing reserve requirements, and/or raising Regulation T

The following section tells you more about these tools.

Opening the Federal Reserve Board's toolbox

The Fed has the authority on behalf of the U.S. government to lend money to banks; it determines the interest rate charged to banks for these loans. You probably remember the chairman of the Fed (currently, Jerome Powell) coming on TV to announce an increase or decrease in the *discount rate* (the rate the Fed charges banks for loans) and what a big deal it was. The rate the Fed charges affects the rates banks charge one another and their public customers. Because banks charge customers higher rates than the Fed charges banks, the Fed policy affects consumers as well (through credit card fees, mortgage loans, auto loans, and so on):

Fed \$ → banks \$ → customers \$

The Fed has a few tools in its arsenal to help control the money supply. (The preceding section explains the effects of tightening and easing the supply.) Here's what you need to understand about these tools for the SIE:

- **Open market operations:** Besides the printing of money, this is the tool the Fed uses most often. *Open market operations* are the buying or selling of U.S. government bonds or U.S. government agency securities to control the money supply. Open market operations are performed by the Federal Open Market Committee (FOMC). If the Fed sells securities, it pulls money out of the banking system; if the Fed purchases securities in the open market, it puts money into the banking system.
- **The discount rate:** This value is the rate that the 12 Federal Reserve Banks charge member banks for loans. If the discount rate increases, the money supply tightens; by contrast, if the discount rate decreases, the money supply eases.
- **Reserve requirement:** The *reserve requirement* is the percentage of customers' money that banks are required to keep on deposit in the form of cash. In line with the theory of supply and demand, if the Fed increases the reserve requirement, banks have less money to lend to customers, so interest rates increase.
- **Regulation T:** *Reg T* is the percentage that investors must pay when purchasing securities on margin (see Chapter 12 for details). Regulation T is currently set at 50 percent, and it doesn't change very often. If the Fed raises the rate, investors have less cash, which tightens the money supply.

HOW THE FED IS SET UP

Here's a quick lesson in government: Congress established the Federal Reserve System in 1913 to stabilize the country's chaotic financial system. The Fed controls our money supply and, therefore, our economy.

The nation is divided into 12 Federal Reserve Districts, each with its own bank. Each bank prints currency to meet the business needs of its district, and each district is distinguished by a letter printed on the face of the bill.

The Federal Reserve Board in Washington is the parent organization that oversees and controls each of the 12 Federal Reserve District Banks. The members of the board, including the chairman, are nominated by the president of the United States, subject to confirmation by the Senate.

Exchange rates

Exchange rates are the rates at which one currency can be converted to another. As you can imagine, exchange rates are constantly changing as the value of currency in different countries appreciates, stays the same, or depreciates. Some investors even speculate in foreign currencies, hoping to be able to purchase a foreign currency when its value is low in the hope that it will appreciate so that they can sell it at a higher value. Certainly, many things can affect the value of a currency, such as a change in a country's social policies, taxing policies, economy, government, and so on.

TIP

You can assume for SIE exam purposes that the value of the U.S. dollar and foreign currency go in opposite directions. The exchange rate is considered to be a *floating rate* because it changes constantly.

U.S. balance of payments

The U.S. *balance of payments* (BoP) is an accounting of the United States' economic transactions with the world over a given period of time (typically quarterly or annually). The balance of payments may show a deficit (more money flowing out of the United States than in) or a credit (more money flowing into the United States than out). As such, the value of our currency (strong or weak dollar) greatly affects our balance of trade and, thus, the U.S. BoP.

If the U.S. dollar is strong in comparison with other currencies, it will be cheaper for Americans to buy foreign goods and services. Thus, more money will likely be going out of the United States.

If the U.S. dollar is weak in comparison with other currencies, it will be cheaper for foreign corporations, governments, individuals, and so on to purchase U.S. goods and services. As a result, more money will be flowing into the United States.

Reading Economic Indicators

Economic indicators are statistics that help show the performance or direction of the economy and help predict the direction of the economy in the near future. The economic indicators are broken into leading indicators, coincident indicators, and lagging indicators.

TIP

For the following section, you should pay attention to the indicators toward the top of each indicator list because those are the ones most likely to be tested.

Leading indicators

Leading indicators are statistics that indicate how the economy is going to (or likely to) do in the future. Leading indicators include

- M2 money supply — the Federal Reserve's estimate of cash people have on hand plus, money available in checking accounts, savings deposits, money market instruments, and so on
- Stock prices
- Fed funds rate (rates that depository institutions [banks and credit unions] charge each other for overnight loans)
- Discount rate (the rate the Fed charges banks for loans)
- Reserve requirements (the percentage of customer deposits that the banks must hold)
- Housing; new construction (building permits)
- Unemployment claims
- Average hours per workweek
- Orders for durable goods (those not for immediate consumption)
- Consumer sentiment
- Yield curves (lines that plot interest rates for bonds of different maturities)

Coincident (coincidental) indicators

Coincident (coincidental) indicators are statistics that indicate how the economy is performing right now. Coincident indicators include

- Industrial production
- Personal income
- GDP
- Number of employees on nonagricultural payrolls and employment levels
- Retail sales

Lagging indicators

Lagging indicators are statistics that mirror leading indicators but reach their peaks and troughs somewhat later. Lagging indicators include

- The prime rate (the rate that banks charge their best customers for loans)
- Outstanding industrial and commercial loans

- Corporate profits
- Ratio of consumer credit to personal income
- Duration of employment — in other words, how long people stay at their jobs before they retire, become unemployed, or move on to a new job
- Unemployment rate
- Ratio of inventories to sales

GDP and GNP: Measuring goods and services

As you can imagine, gross national product (GNP) and gross domestic product (GDP) are closely related terms. When you are taking the SIE exam, you should understand the difference between the two. Because the SIE is mainly taken by U.S. residents, I focus on the GDP and GNP in U.S. terms. Certainly, the better the GNP and the GDP, analysts would see that as a good sign.

- **GDP:** To cut the information down to what you need to know for the SIE exam, think of the GDP as the total of all goods produced and all services provided by the United States in a one-year period.
- **GNP:** GNP includes the GDP plus the investments made by U.S. businesses and residents inside and outside the United States. However, the GNP doesn't count income earned by foreign businesses or residents in the United States. It also excludes products from overseas firms manufactured in the United States.

Because inflation occurs over time, the GDP and GNP are measured in *constant dollars,* which includes the cost of inflation. This is necessary to help determine whether the economy is expanding or contracting from year to year.

How Economic Factors Affect Securities

Economic factors affect securities differently. When taking the SIE exam, you are expected to understand how the economy affects *cyclical, defensive,* and *growth* companies.

Cyclical

A *cyclical* corporation is one in which the performance depends on the economy. If the economy is expanding, cyclical companies will do well. If the economy is contracting, cyclical companies will be heavily affected. Some examples of cyclical corporations are household appliances, automobile companies, tourism, construction, and manufacturing. Obviously, when the economy is not doing well, people will wait to buy that new refrigerator or that new car, wait to go on vacation, and so on. As such, cyclical companies' stock will likely decrease, and these companies may have problems paying interest on their debt securities.

Countercyclical stocks such as fast food, auto parts, discount retailers, and other investments such as precious metals may actually do better as the economy gets worse.

Defensive

Defensive corporations are ones whose sales remain relatively stable no matter how the economy performs. The corporations offer products or services that people will purchase even in a contracting economy. Examples of defensive companies include utilities, food, clothing, alcohol, tobacco, and cosmetics.

Growth

Growth companies, such as technology companies, are ones that are growing at a more rapid pace than comparable companies or the market as a whole. Growth companies are typically the newer companies and are more likely to do extremely well during periods in which the economy is expanding, but they underperform other stocks, such as value stocks, when the economy is contracting and the market is bearish. Because growth companies invest a lot of money back into their business, they typically don't pay much (or anything) in dividends.

Your Principal Economic Theory Primer

Economists follow principal economic theories. These theories are important because the administration that's in power typically nominates people to help run our economy who share their same beliefs. Economists typically believe in one of these theories, and there has been much debate about which one is best for the economy. Fortunately for you, you don't have to figure that out; you only need to understand the basics of each theory to help you pass the SIE exam. This list describes the three main economic theories that economists usually follow.

- **Keynesian (demand side):** The basis of the Keynesian theory is that government intervention through fiscal policy is good to help stimulate consumer demand for goods and services. Keynesian policy typically involves raising taxes, enabling deficit spending, borrowing money, and printing currency.
- **Supply side (Reaganomics):** Supply-side economics is the theory that the government should remain relatively inactive and the economy will grow by itself. According to this theory, the less regulation and the lower the taxes are, the more businesses will reinvest and the more people they will hire.
- **Monetarist:** Monetarists believe that the economy's performance is largely determined by the money supply (controlled by the Fed). The money supply can be used to fight inflation or to stimulate the economy.

Testing Your Knowledge

This chapter, probably more than any other one, requires you to think about what happens. What happens, for example, if the Fed raises interest rates? Or what happens if a company's liabilities increase? This type of analysis is important because it helps analysts and investors decide which securities to buy or sell.

As with other chapter questions, I try to give you a wide variety of potential questions you may see on the exam.

Practice questions

1. Which of the following are defensive industries?

- **I.** Utilities
- **II.** Tourism
- **III.** Household appliances
- **IV.** Food

- **(A)** I and IV
- **(B)** II, III, and IV
- **(C)** II and III
- **(D)** I, II, III, and IV

2. All of the following are examined by a fundamental analyst EXCEPT

- **(A)** earnings per share
- **(B)** balance sheets
- **(C)** the industry
- **(D)** timing

3. All of the following are bearish positions EXCEPT

- **(A)** buying inverse ETFs
- **(B)** selling uncovered call options
- **(C)** selling short
- **(D)** selling naked put options

4. Which of the following best describes the discount rate?

- **(A)** The interest rate that banks charge one another for overnight loans
- **(B)** The interest rate that the banks charge their best customers for loans
- **(C)** The interest rate that the Fed charges banks for loans
- **(D)** The interest rates charged in margin accounts

5. Melissa wants to invest in a retirement plan that protects against purchasing power risk. Which of the following would be the most suitable investment?

- **(A)** Municipal bonds
- **(B)** Common stock
- **(C)** Variable annuities
- **(D)** Fixed annuities

6. Looking at the stages of the business cycle, if the economy is expanding, what stages would you expect to follow, in order from first to last?

I. Trough
II. Peak
III. Contraction

(A) I, II, III
(B) II, III, I
(C) III, II, I
(D) I, III, II

7. Which TWO of the following actions may the Fed take to ease the money supply?

I. Purchase T-bills
II. Sell T-bills
III. Increase reserve requirements
IV. Decrease reserve requirements

(A) I and III
(B) I and IV
(C) II and III
(D) II and IV

8. All of the following are leading indicators EXCEPT

(A) the prime rate
(B) M2 money supply
(C) the discount rate
(D) stock prices

9. Which of the following is measured in constant dollars?

(A) The Fed funds rate
(B) The M2 money supply
(C) The prime rate
(D) GDP

10. Proponents of the Dow Theory look at

(A) the Dow Jones Composite Average and the DJIA
(B) the Dow Jones Utility Average and the Dow Jones Transportation Average
(C) the DJIA and the Dow Jones Transportation Average
(D) the Dow Jones Composite Average and the Dow Jones Utility Average

11. Zero-coupon bonds have no

(A) reinvestment risk
(B) purchasing power risk
(C) market risk
(D) interest risk

12. All of the following can be found on a corporation's balance sheet EXCEPT

(A) fixed assets
(B) long-term liabilities
(C) retained earnings
(D) net income

13. Which of the following is a way an investor can diversify investments?

(A) Geographically
(B) Investing in stocks from different sectors
(C) Purchasing bonds with different maturity dates
(D) All of the above

14. A holder of which of the following securities is the most exposed to inflationary risk?

(A) U.S. Treasury bonds with ten years until maturity
(B) U.S. Treasury bills with one year until maturity
(C) Stocks issued by a utility company
(D) Pharmaceutical stocks

15. A registered representative from Missed Again Securities is reviewing one of their client's portfolios. They see that the client has 35 percent invested in TUV Energy Company, 15 percent invested in ETFs tracking the S&P 500, 35 percent in LMN Healthcare Company, 15 percent in U.S. government securities, and 10 percent in money market funds. Which of the following risks are built into this portfolio?

(A) Liquidity risk
(B) Political risk
(C) Credit risk
(D) Nonsystematic risk

16. Which of the following risks apply to both domestic and foreign debt securities?

(A) Regulatory
(B) Political
(C) Interest rate
(D) All of the above

17. The greatest investment risk that an investor would face when purchasing a variable annuity is

(A) interest rate risk
(B) market risk
(C) purchasing power risk
(D) credit risk

Answers and explanations

1. **A.** Defensive industries are ones that are pretty much resistant to economic downturns. These include utilities, food, clothing, alcohol, tobacco, cosmetics, and so on. Tourism and household appliance–related companies suffer during economic downturns and, therefore, are considered to be cyclical companies.

2. **D.** Fundamental analysts look at and compare companies. They do an in-depth analysis to help professionals decide which companies would be good or bad investments. So fundamental analysts help decide what to buy, whereas technical analysts look at the market and help professionals decide when to buy (timing).

3. **D.** Bearish strategies are ones in which you want the price of the security to decrease. Bearish strategies include selling short, buying bearish funds, buying inverse exchange-traded funds, selling uncovered (naked) call options, and so on. Selling naked (uncovered) put options is actually a bullish strategy because the investor who holds that position wants the price of the underlying security to increase.

4. **C.** The discount rate is the rate that the Fed charges banks for loans. These loans are typically overnight loans taken by banks to help them meet their reserve requirements.

5. **C.** You can cross off (A) and (B) right away because stocks and bonds are not retirement plans. Out of the two annuity plans, variable annuities do not have purchasing power (inflation) risk because the return is based on the performance of the securities held in the separate account. Ideally, the securities will outperform the rate of inflation.

6. **B.** If the economy is expanding, you would expect it to hit a peak at some point. After reaching the peak, the economy would start to contract until hitting the lowest point, which is the trough. After that, you would expect it to start all over again.

7. **B.** The Fed has a few tools to help ease or tighten the money supply including open market operations, adjusting the discount rate, changing reserve requirements, and raising or lowering Regulation T requirements. Of the choices given, the two that would ease (add money to) the money supply would be inserting money into the system by purchasing U.S. government securities like T-bills and lowering reserve requirements (the percentage of customer deposits banks have to hold each night).

8. **A.** Leading indicators are those that help give an indication of how the economy is going to do. The main leading indicators are M2 money supply, stock prices, the Fed funds rate, and the discount rate. The prime rate — the rate that banks charge their best customers for loans — is a lagging indicator. Lagging indicators go the same direction as the leading indicators but arrive a little later.

9. **D.** Both the GDP and GNP are measured in constant dollars. When measuring in constant dollars, they are factoring inflation into the equation to see whether the economy is actually expanding or contracting year to year.

10. **C.** Proponents of the Dow Theory logically believe that major market trends are confirmed if both the DJIA and the Dow Jones Transportation Average are going the same direction. In other words, companies not only have to be making a lot of goods, but also have to be transporting those goods to be considered a positive sign.

11. **A.** Reinvestment risk is the additional risk taken with interest or dividends received. Zero-coupon bonds are issued at a discount and mature at par value. These bonds do not make interest payments along the way, so there is nothing to reinvest.

12. **D.** The balance sheet is just a snapshot of the net worth (stockholders' equity) of a company. The balance sheet includes the company's assets, liabilities, and stockholders' equity. Net income is derived from a company's income statement. The income statement looks at a company's income minus expenses.

13. **D.** Diversification helps investors mitigate investment risk. All the choices listed are ways investors can lessen risk. When dealing with clients, you should always help them diversify their portfolio. It's the old "don't put all of your eggs in one basket" theory.

14. **A.** Inflationary risk (purchasing power risk) is the risk that inflation can pose to a security or portfolio over time. Treasury bonds (T bonds) have a longer maturity, and if inflation kicks in, the Fed is likely to raise interest rates. If interest rates increase, outstanding bond prices decrease. Longer-term bonds are affected more than shorter-term bonds, like Treasury bills.

15. **D.** This investor has 70 percent of their portfolio invested in two different companies (35 percent in TUV and 35 percent in LMN). Therefore, this customer has nonsystematic (unsystematic or diversifiable) risk built into their portfolio. Nonsystematic risk is the risk that a particular company or industry might do poorly and bring the value of the portfolio down substantially. Obviously, you would look at this portfolio and say that the investments in U.S. government securities, money market funds, and ETFs provide a certain degree of safety. However, those investments are only 30 percent of their portfolio. Therefore, you would likely recommend that the other 70 percent should be spread out to more than two companies.

16. **D.** All of the choices listed can apply to both domestic and foreign debt securities. Regulatory risk is the risk that law changes will affect the market. Political risk is the risk that politics in a particular country could negatively affect securities. Interest rate (money rate) risk is the risk that bond prices will decline due to increasing interest rates.

17. **B.** The greatest investment risk that an investor will face when purchasing a variable annuity is market risk. Variable annuities have a separate account where an investor chooses the securities held. The securities held in the account are subject to market risk, which is the risk that the securities can decline due to negative market conditions.